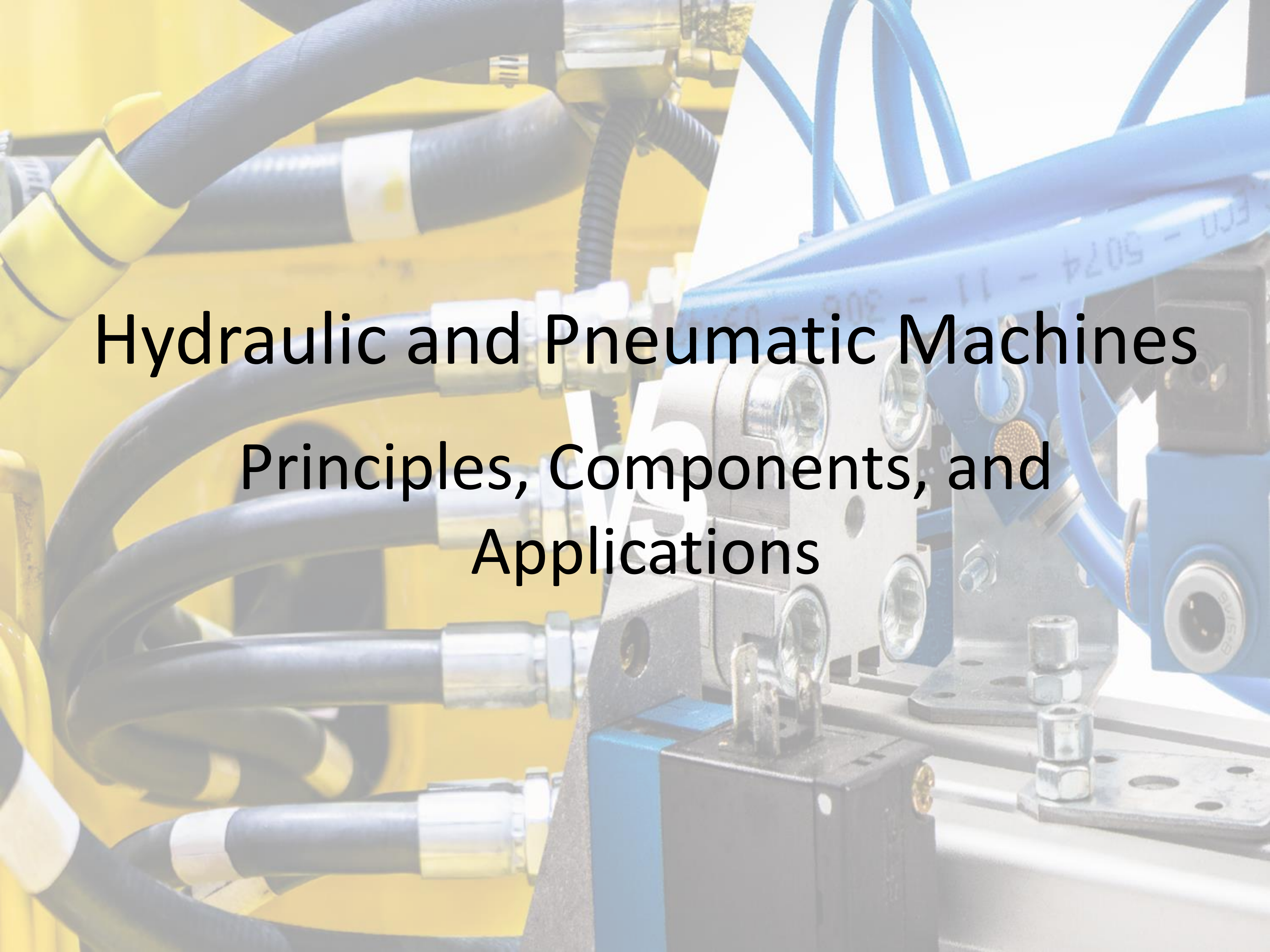
Industrial Electronics

BECS 44453 / ELEC 44163

Muditha Prasanna



Hydraulic and Pneumatic Machines

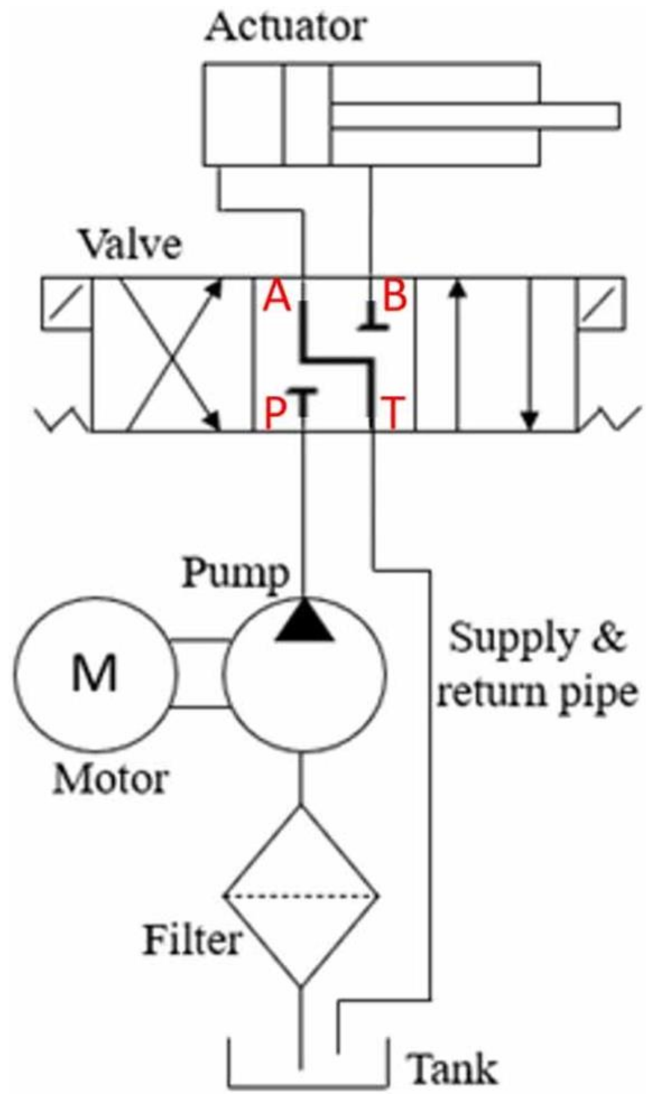
Principles, Components, and Applications

Basic Principles

- **Hydraulic Systems**
  - Use of incompressible liquids (oil/water)
  - Pascal’s Law: Pressure applied to a confined fluid is transmitted equally in all directions
- **Pneumatic Systems**
  - Use of compressed gases (air)
  - Gas compressibility allows for different control mechanisms

Components of Hydraulic Systems

- Hydraulic Pump
- Hydraulic Cylinder
- Hydraulic Motor
- Valves (Control, Relief)
- Reservoir
- Filters



Hydraulic Pump

Centrifugal Pump

Diaphragm Pump

Electromagnetic Pump

Jet Pump

Mixed Flow Pump

Gear Pump

Piston Pump

Progressive Cavity Pump

Plunger Pump

Screw Pump

Axial Flow Pump

Valveless Pump

Hydraulic Pump

Converts mechanical energy into hydraulic energy by pressurizing the hydraulic fluid.

Types of Hydraulic Pumps:

- **Gear Pump:**
- **Vane Pump:**
- **Piston Pump:**

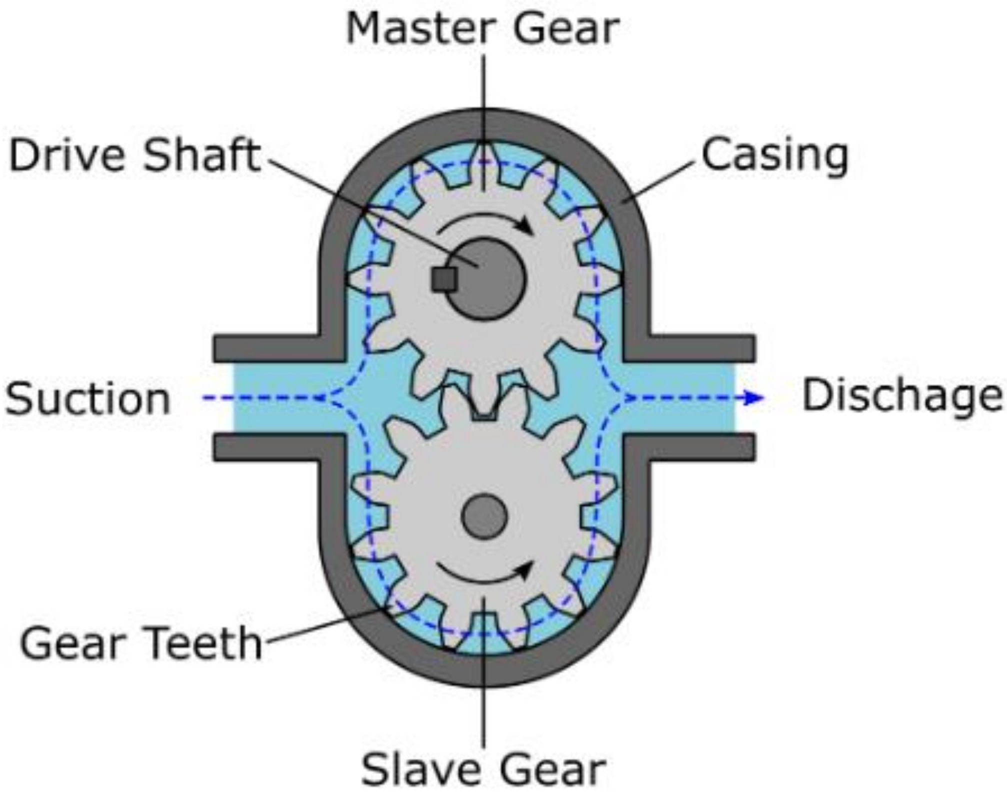
Working Principle:

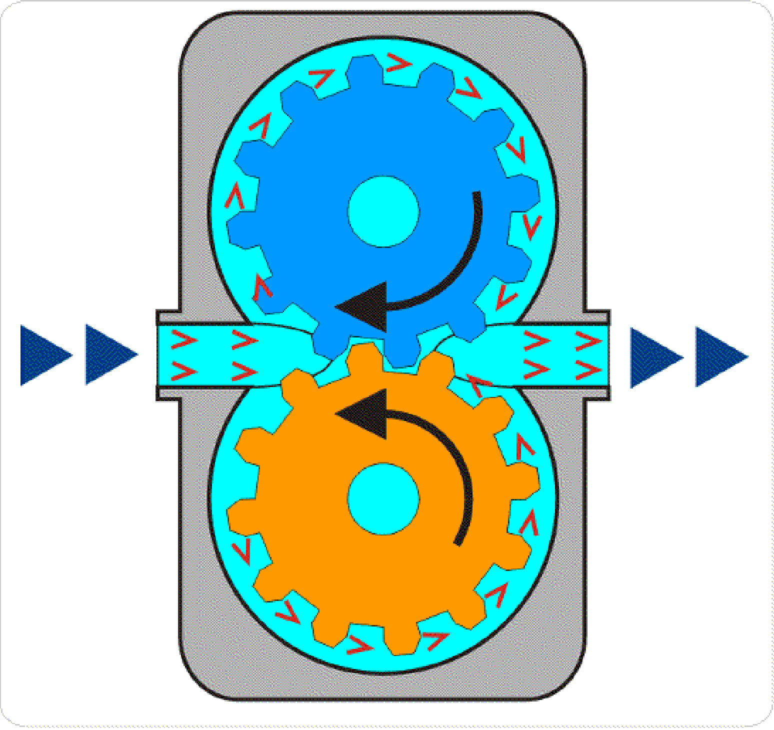
- Mechanical input (from an electric motor or engine) drives the pump.
- Fluid is drawn into the pump chamber and then forced out under pressure.

Key Features:

- Flow rate determines the volume of fluid delivered.
- Pressure depends on system resistance and pump design.

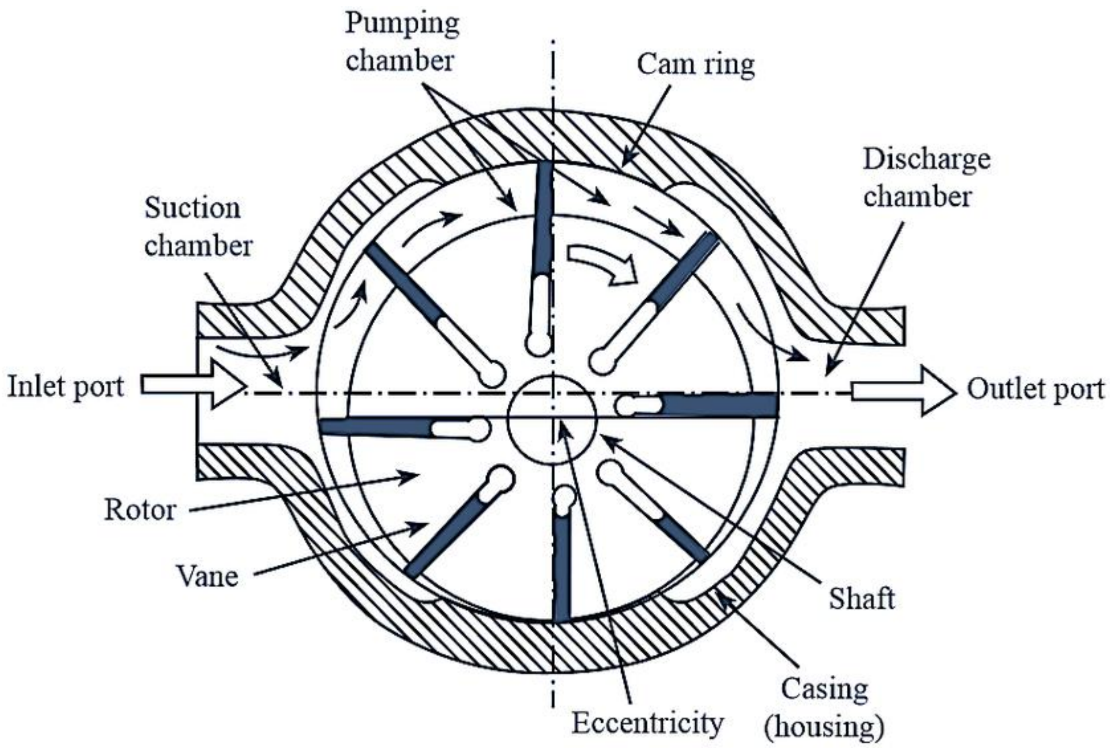
Gear Pump

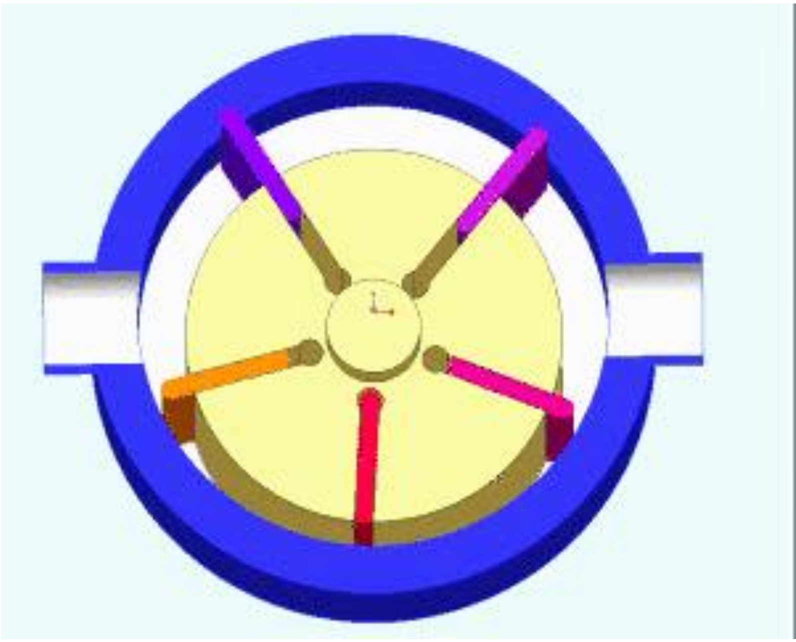




Uses meshing gears to pump fluid; simple and inexpensive.

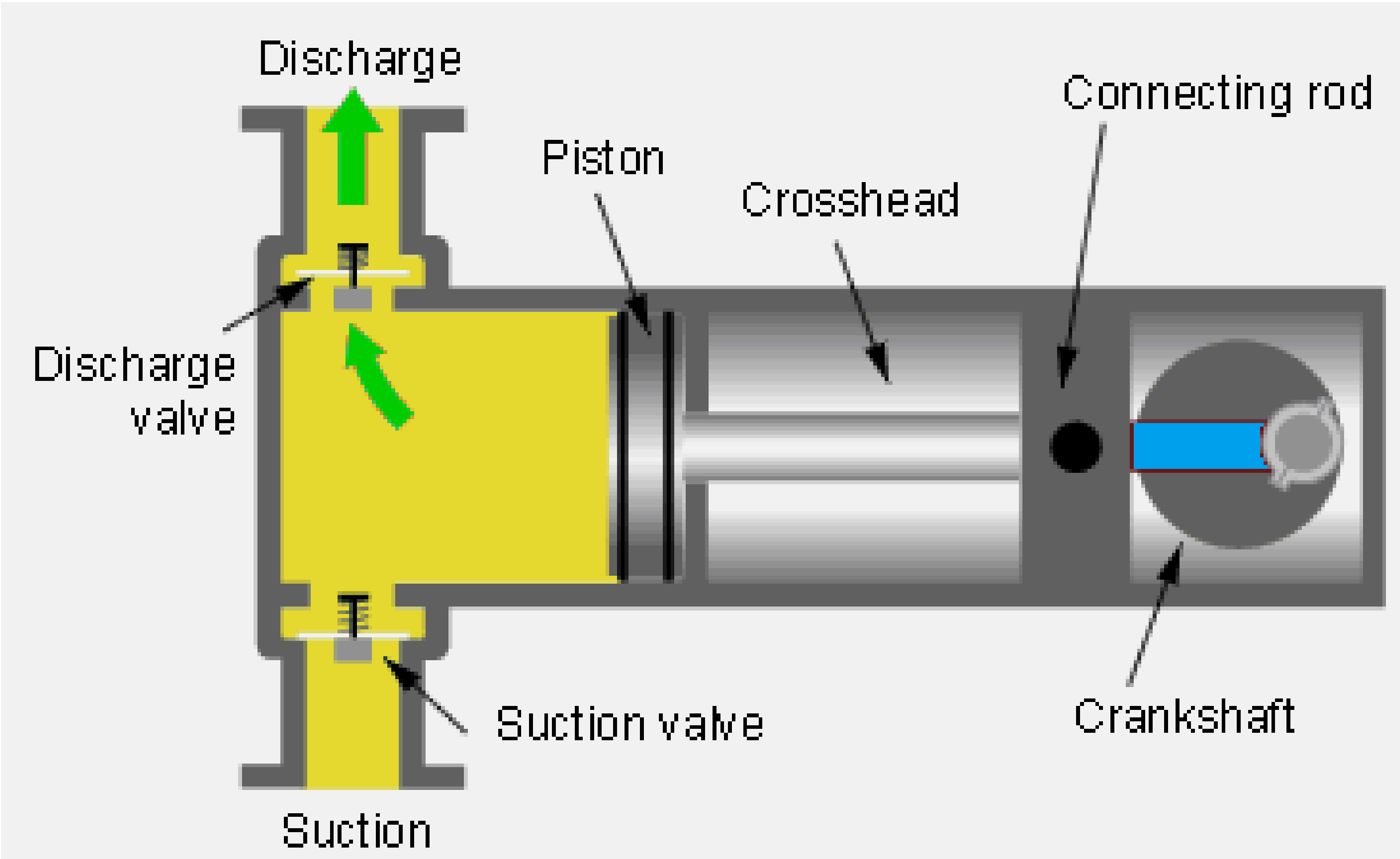
Vane Pump





Uses a rotating vane within a chamber; offers smooth flow.

Piston Pump



Uses pistons reciprocating within cylinders; high pressure and efficiency.

Hydraulic Cylinder

Converts hydraulic energy into linear mechanical motion.

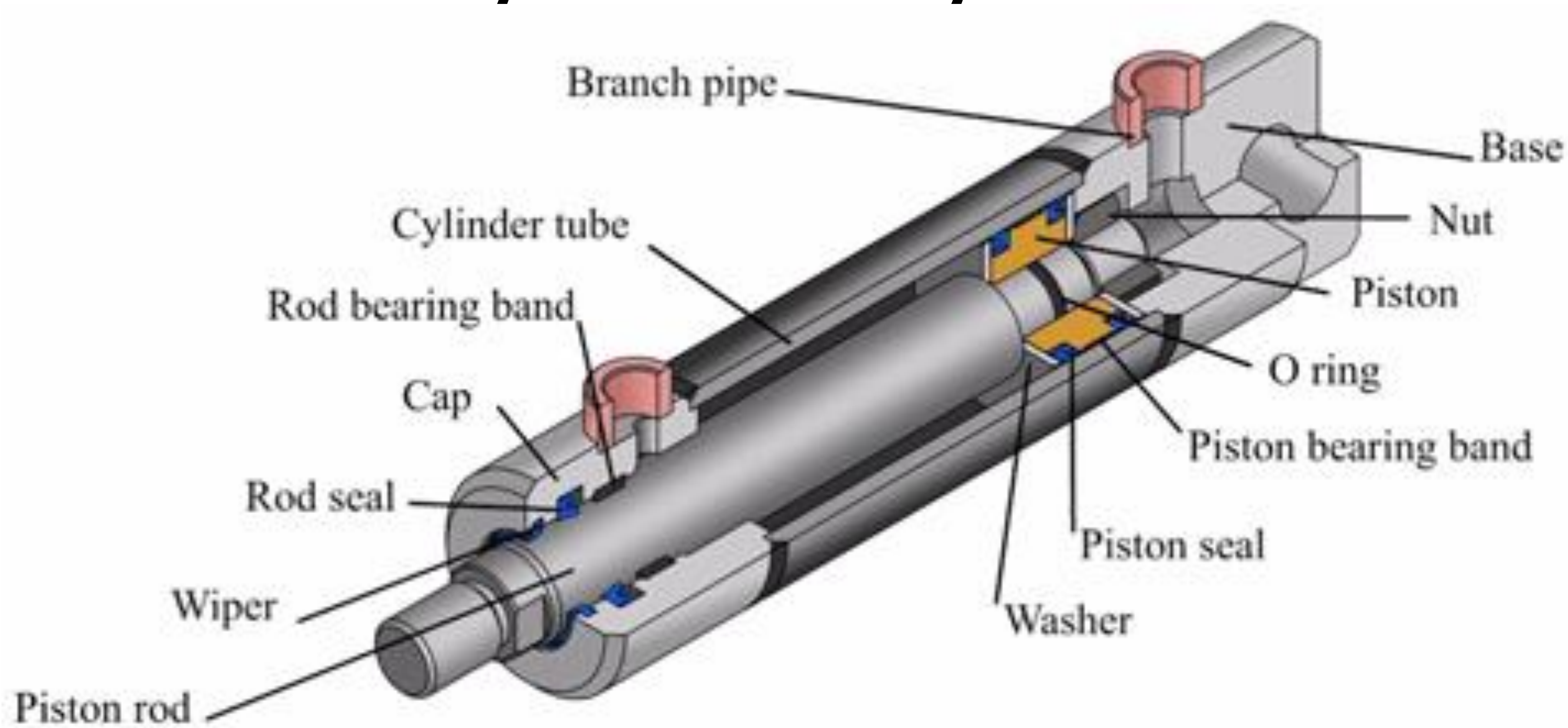
Working Principle:

- Hydraulic fluid enters one side of the piston, exerting force.
- The piston moves linearly, performing work such as lifting or pushing.

Types of Cylinders:

- **Single-Acting**
- **Double-Acting**

Hydraulic Cylinder



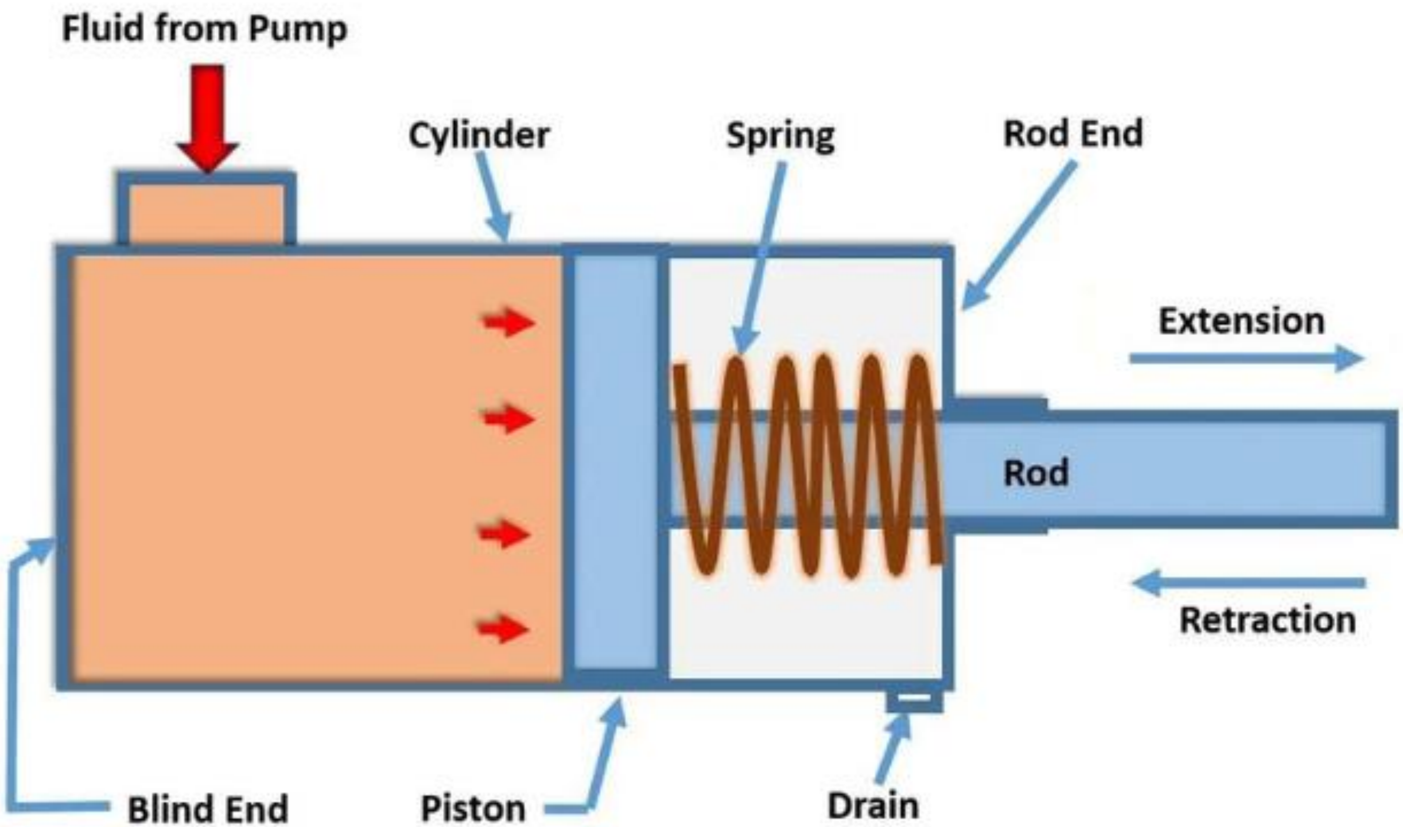
Main Parts:

Cylinder Barrel: The main body where piston moves.

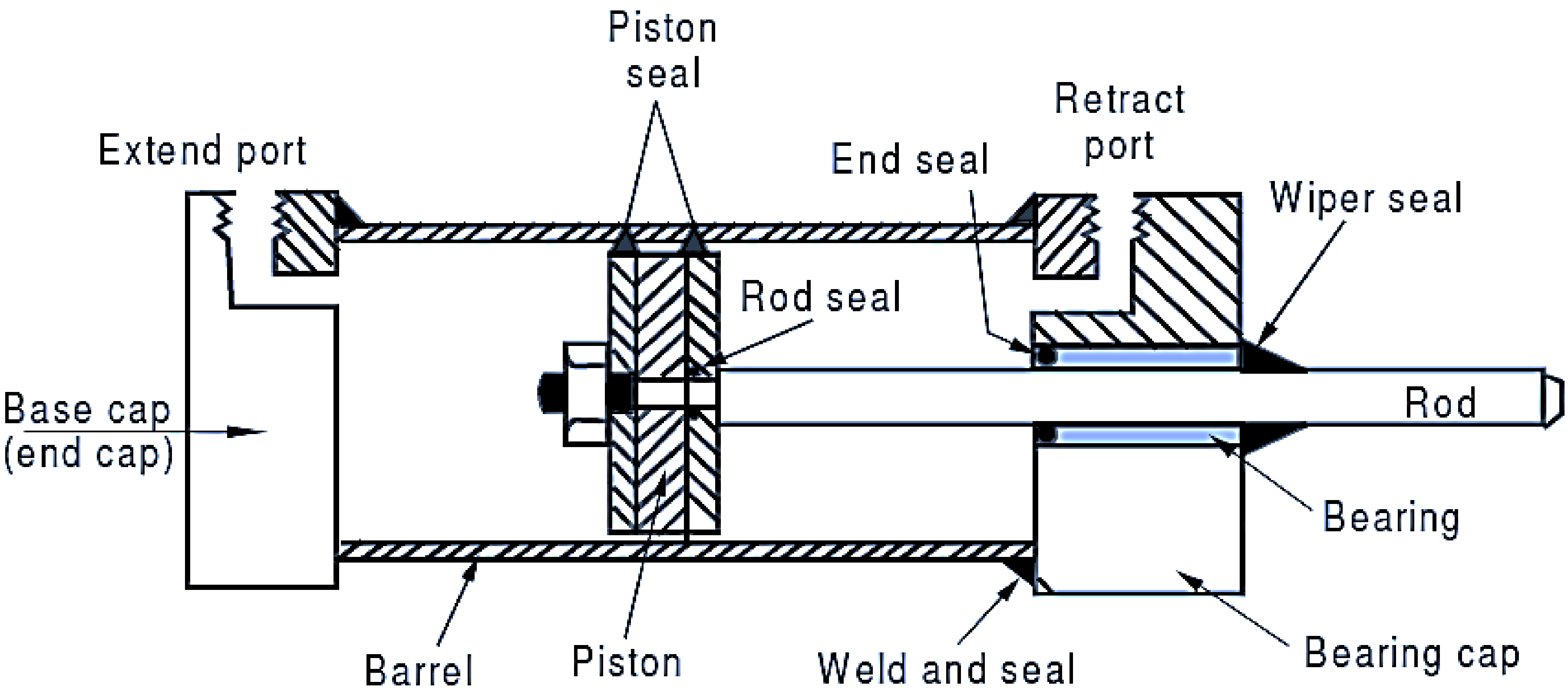
Piston and Piston Rod: Moves inside the barrel; piston divides the chamber into two parts.

End Caps: Seal the ends of the cylinder.

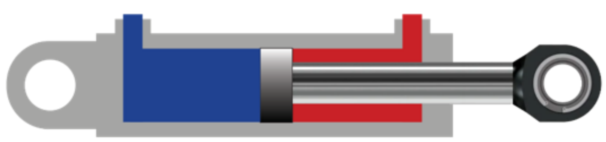
Single-Acting: Power is supplied in one direction; spring or external force returns the piston



Double-Acting: Power can be applied in both directions for extending and retracting.



	SINGLE-ACTING	DOUBLE-ACTING
Retraction Method	Return spring, load, or gravity	Hydraulic return
Connection Ports	1	2
Advantages	✓ Simplicity ✓ Requires just 1 hose per cylinder ✓ Economical ✓ Easy to maintain	✓ Precise control ✓ Faster retraction ✓ Suits repetitive actions ✓ Applies push and pull forces
Disadvantages	× Pushes in just one direction. × Less controlled retraction. × Springs can wear out.	× More complex × Needs 2 x hoses × Cost × Requires pump with a 4-way valve
Useful for	• Simple lifting jobs. • Light industrial and commercial applications. • When fast retraction is not essential.	• Repetitive actions. e.g. jack and crib applications. • Presses, or when the cylinder is upside down. • When you want both pushing and pulling forces. • When very long hoses are required. • A controlled retraction time.



Hydraulic Motor

Converts hydraulic energy into rotary mechanical motion.

Working Principle:

- Pressurized fluid enters the motor, causing internal parts to rotate.
- Rotational force is transmitted to machinery or wheels.

Applications:

- Driving conveyor belts, winches, or steering systems.

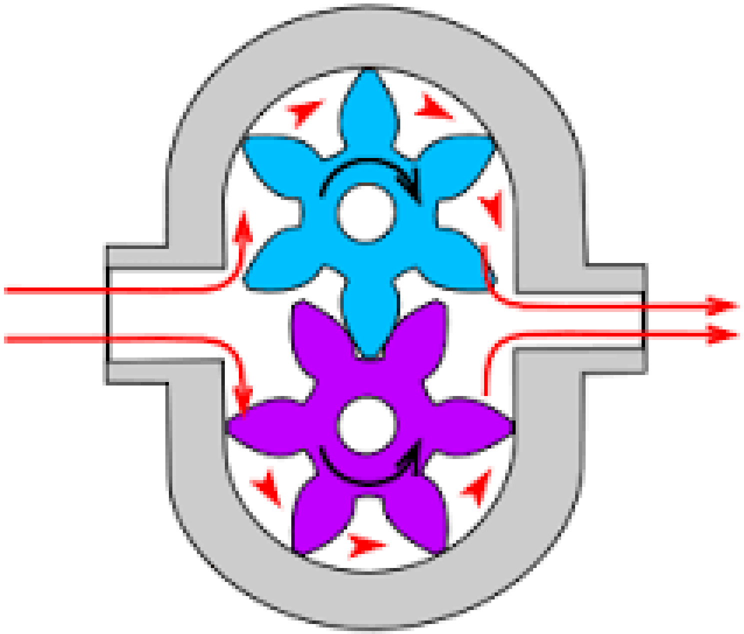
Hydraulic Motor

Types:

- Gear Motor
- Vane Motor
- Piston Motor

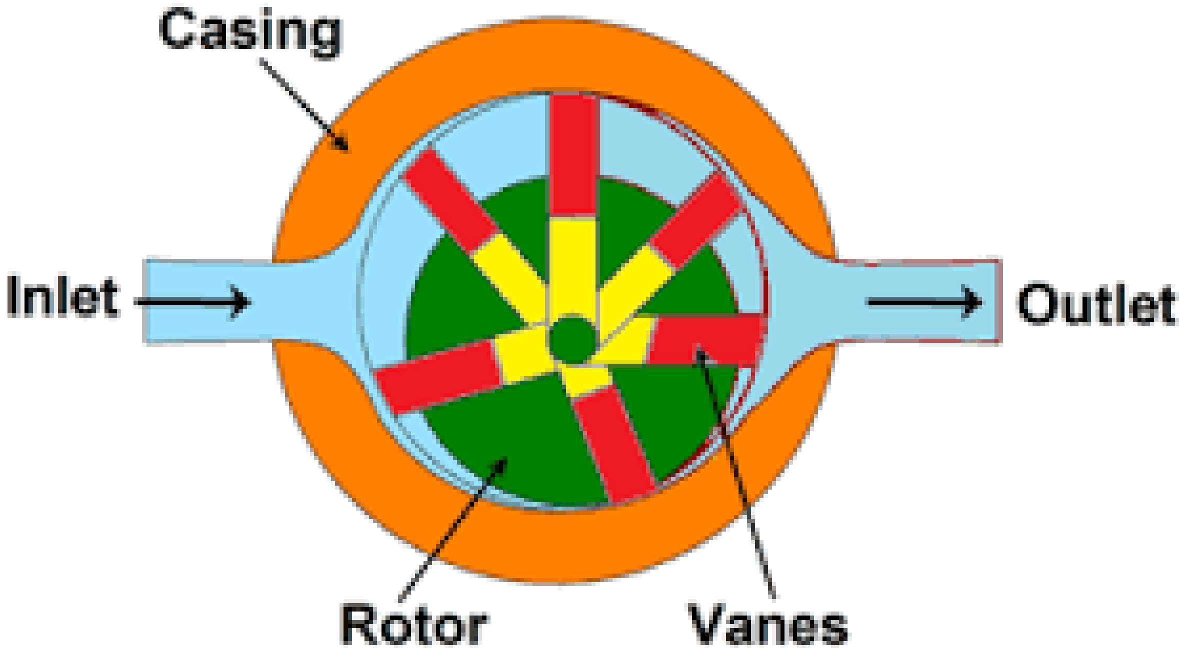
Gear Motor:

Uses gears for rotary motion; simple and compact.



Vane Motor

Uses vanes that slide in and out of a rotor.



Piston Motor

Uses pistons in a cylinder for high torque.

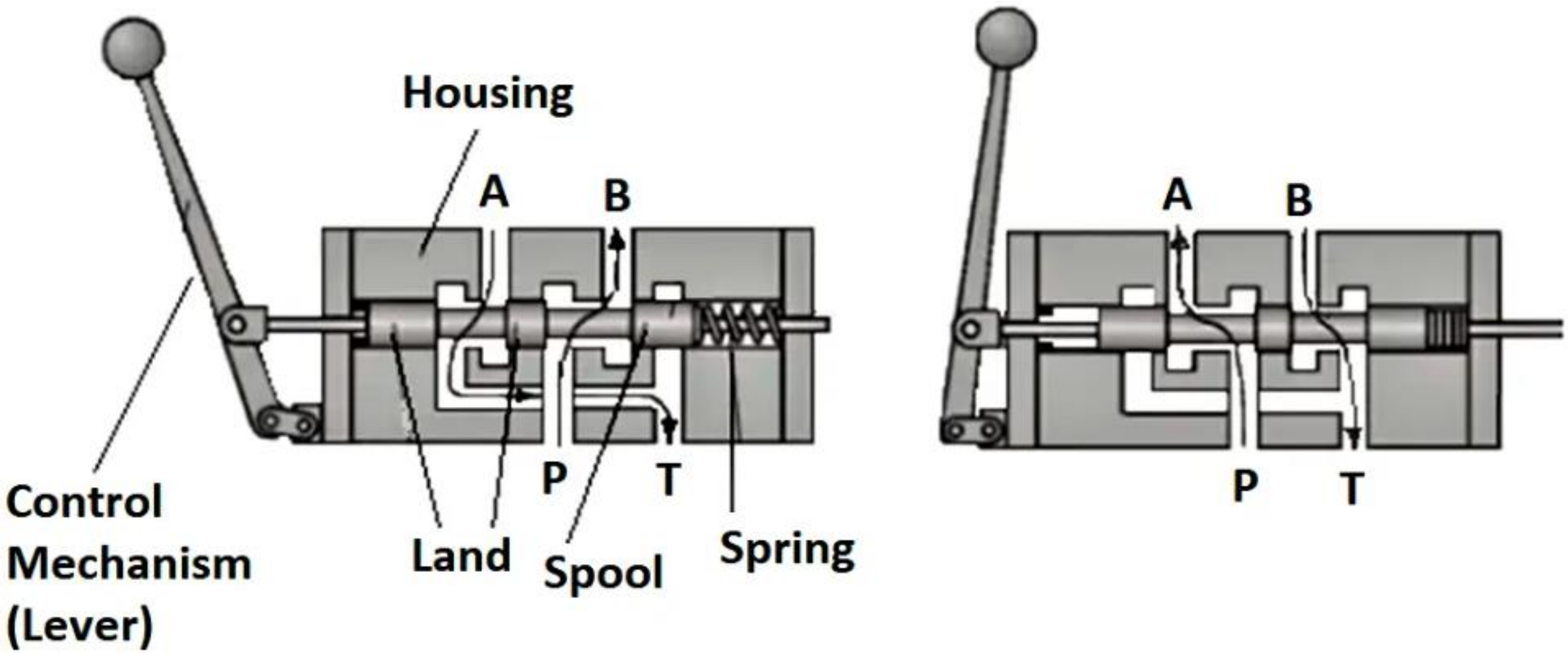
Valves (Control and Relief)

Control Valves:

- Regulate flow direction, pressure, or flow rate.
- Types:
  - **Directional Control Valves (DCV),**
  - **Flow Control Valves,**
  - **Check Valves.**

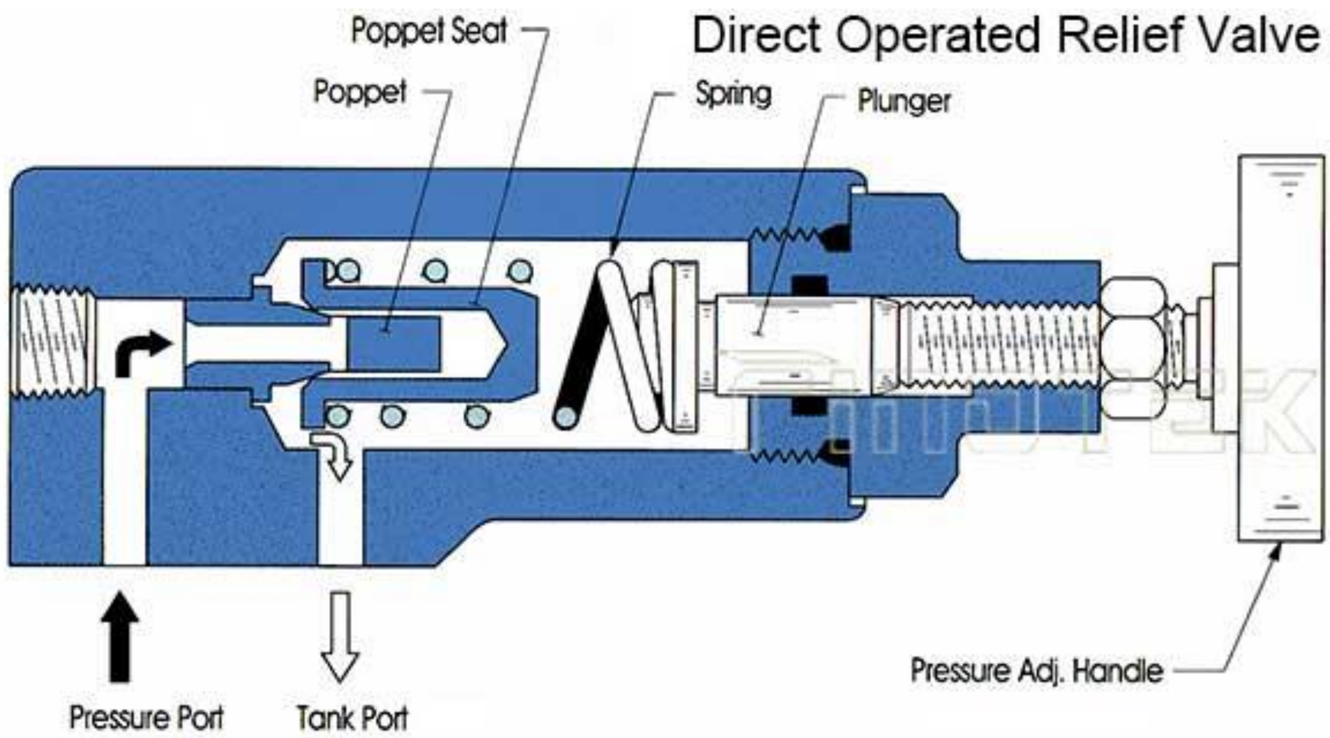
Directional Control Valve (DCV):

Controls the path of fluid flow.



Relief Valve:

- Protects the system from excessive pressure.
- Opens automatically when pressure exceeds set level, allowing fluid to return to the reservoir.



Control Valves:

- Valves are operated manually, mechanically, or electrically (solenoid valves).
- They direct fluid to appropriate components for desired motion.

Reservoir

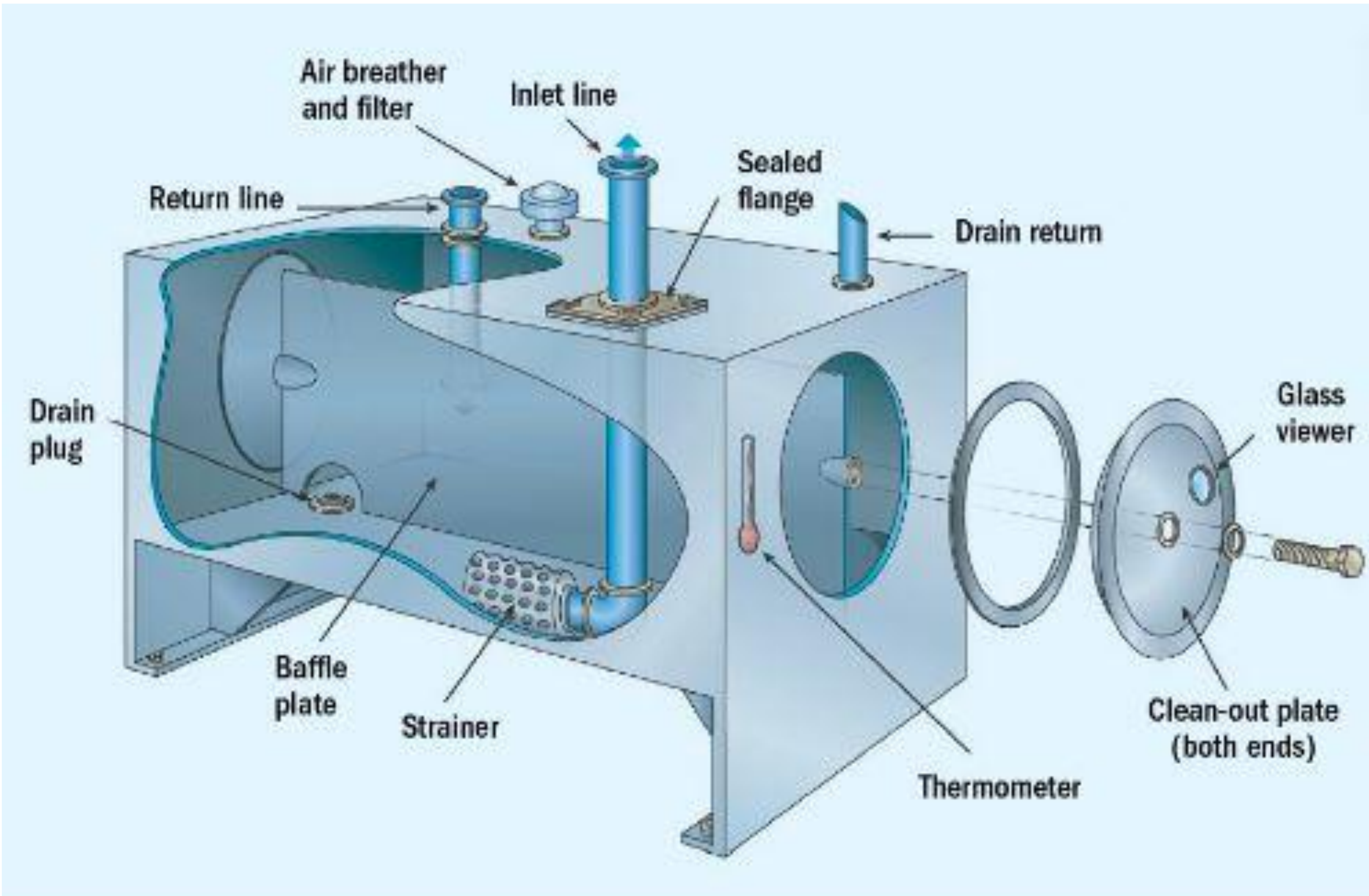
Stores hydraulic fluid and maintains system pressure.

Functions:

- Supplies fluid to the pump.
- Allows air bubbles and contaminants to separate from the fluid.
- Provides a reservoir for fluid expansion and contraction.

Design Features:

- Usually a tank with an inlet (from pump) and outlet (to system).
- Includes breather vents, filters, and level indicators.



Reservoir

Maintenance:

- Regularly check fluid levels.
- Clean and replace filters as needed.

Filters

Purpose:

Remove dirt, debris, and contaminants from hydraulic fluid to prevent damage and wear.

Parameters of the filter

- **Beta ratio**:

  Beta (β)Ratio = (Particle count in the upstream oil) ÷ (Particle count in the downstream oil)

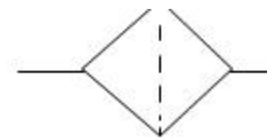
- **Filter efficiency**

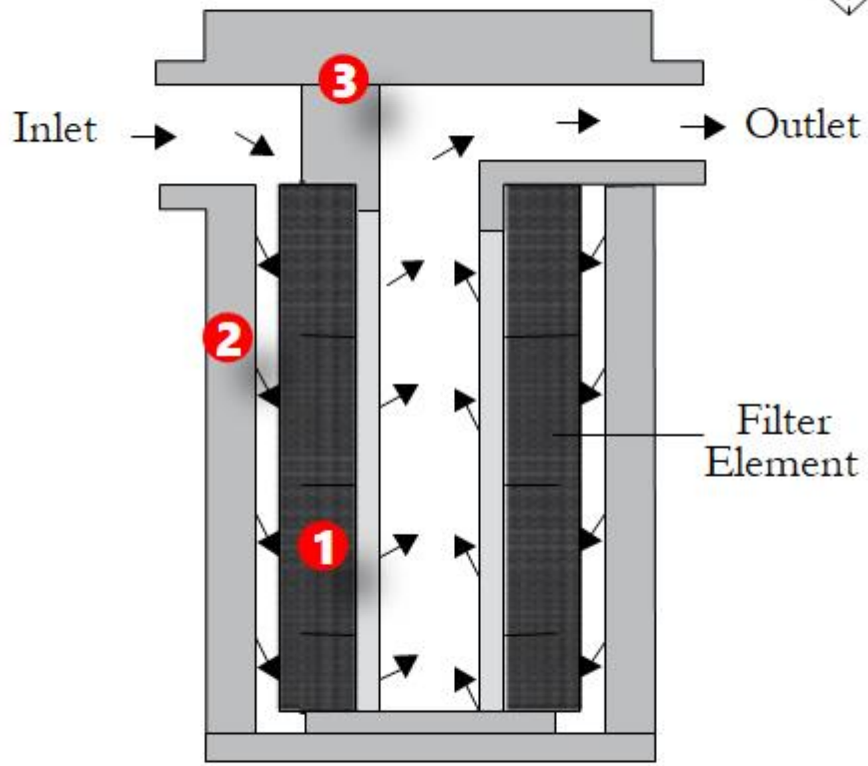
  Efficiency (x) = [1- (1/ β)] x 100

- **micron rating**:

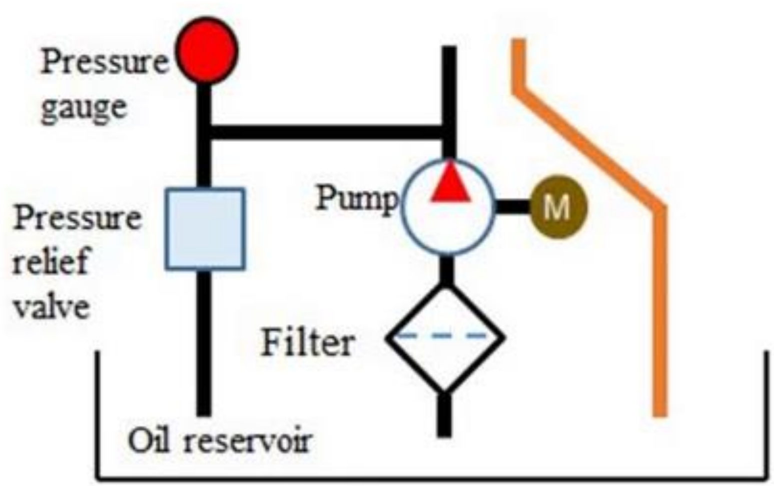
  The smallest micron size of particles it can capture in a specified quantity

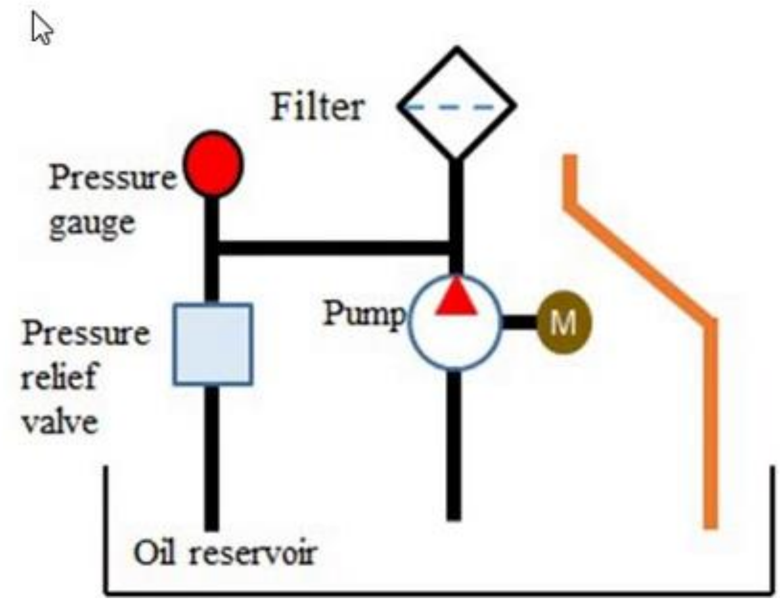
Intake Filter: Located at the pump inlet, filters incoming fluid.



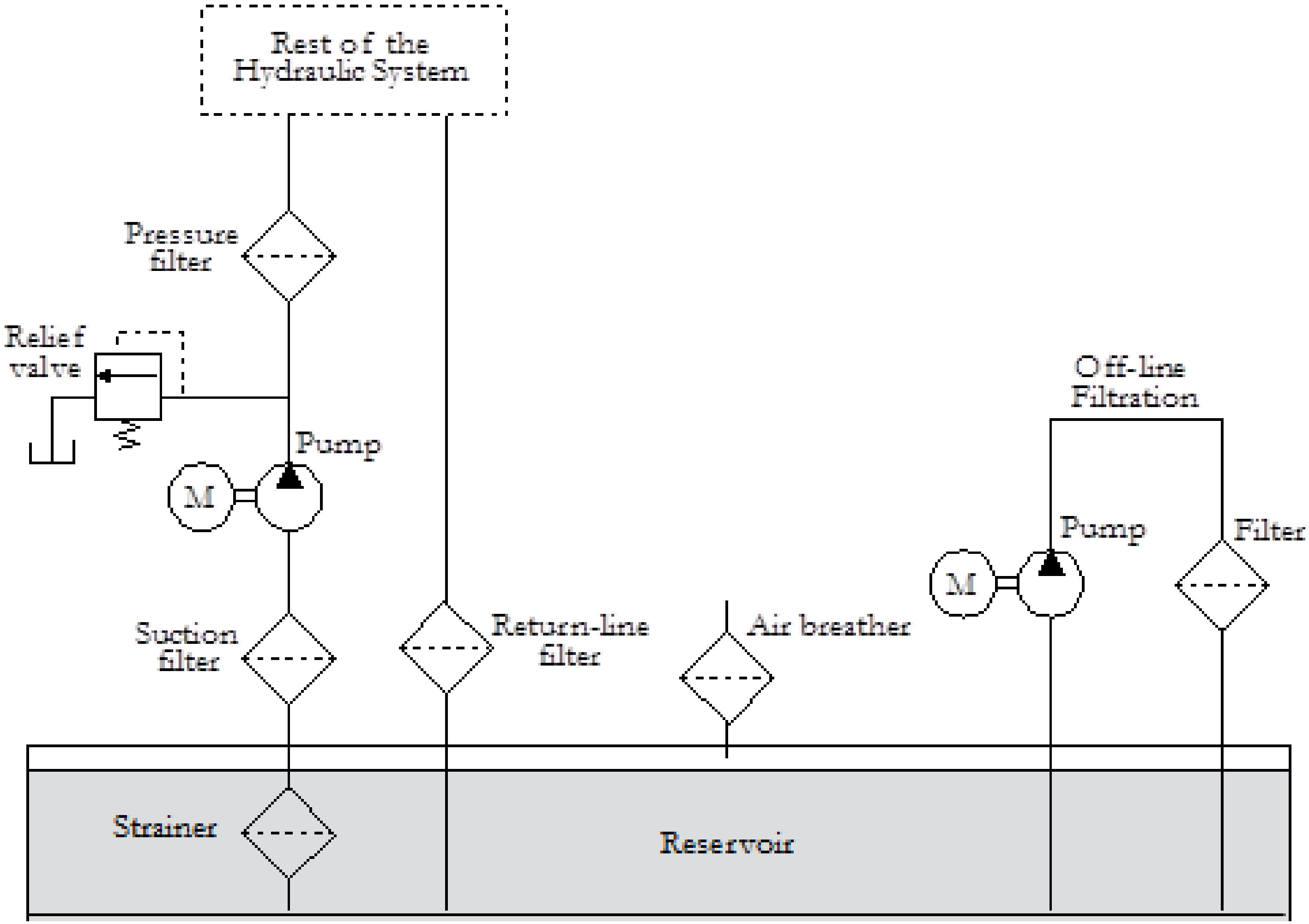


In-line Filter: Installed along fluid lines for continuous filtration.





Return Filter: Cleans fluid returning to the reservoir.

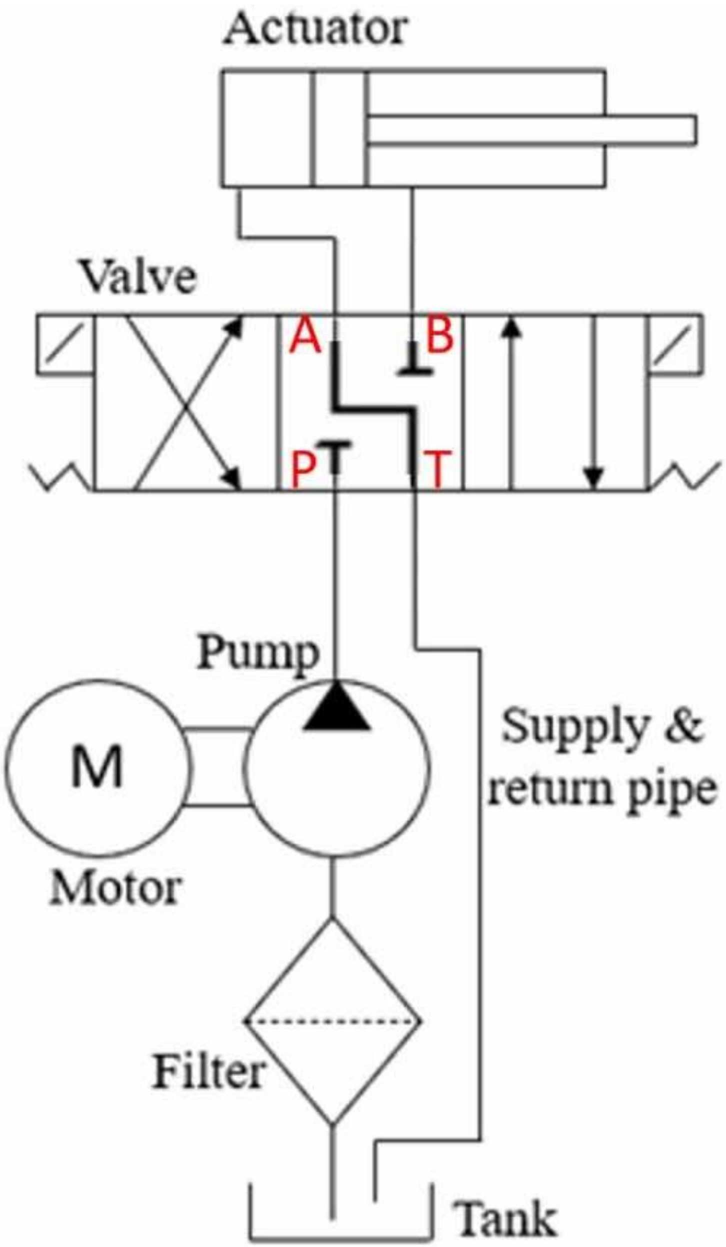


Importance of Filtration:

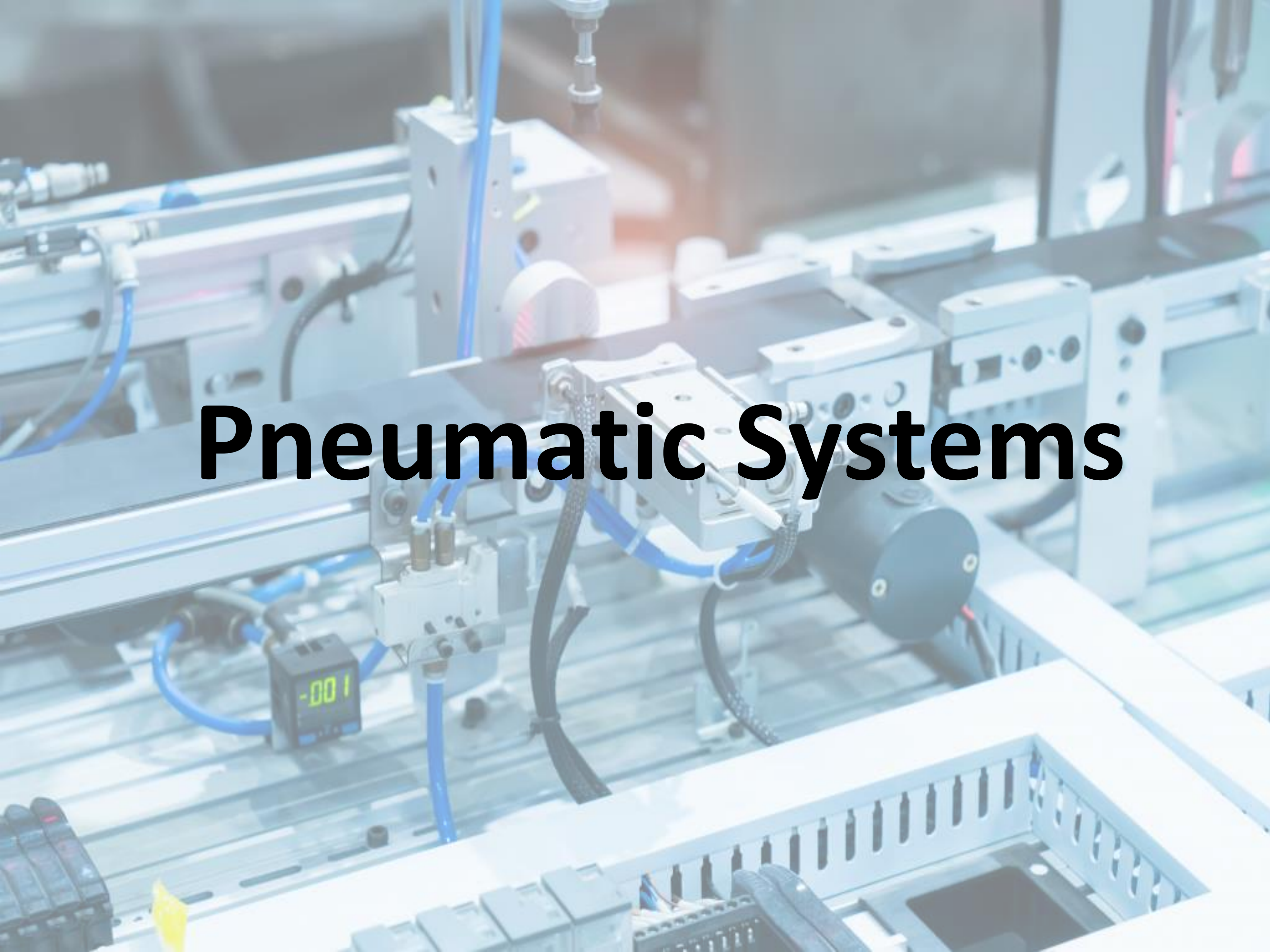
- Ensures smooth operation and longevity of system components.
- Prevents clogging, wear, and system failure.

Maintenance:

- Regular inspection and replacement based on operating conditions.


Actuator
Valve
A
B
P
T
Pump
M
Motor
Supply &
return pipe
Filter
Tank

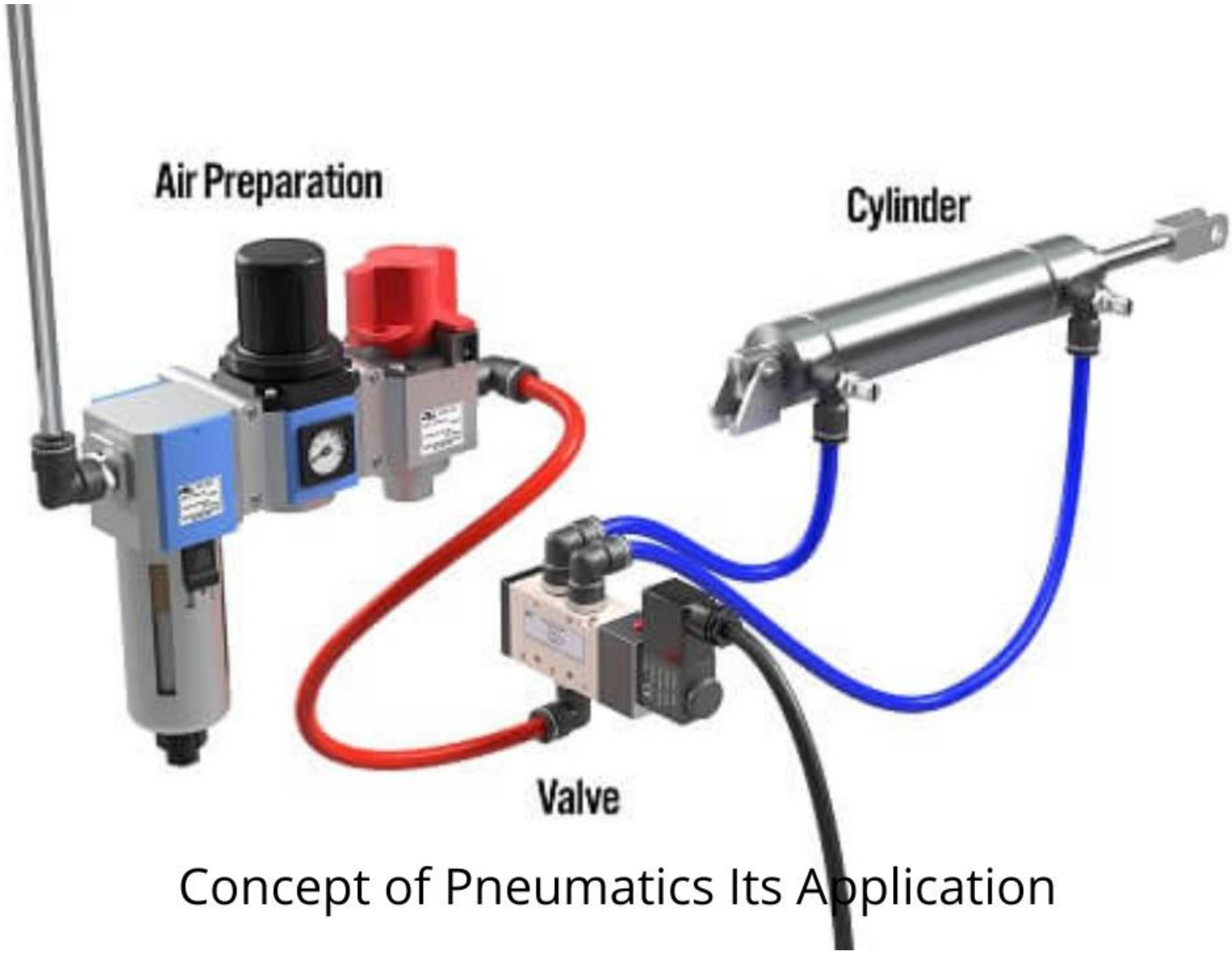
Pneumatic Systems



Introduction to Pneumatic Systems

Pneumatic systems utilize compressed air to transmit and control energy for performing various mechanical tasks.

- **Applications:** Used in manufacturing industries, automation, robotics, aircraft systems, and tools.
- **Advantages:**
  - Clean and dry operation, suitable for food and pharmaceutical industries.
  - Safe (no risk of electric shocks or fires).
  - Fast and precise control.
  - Simple and economical to maintain



Concept of Pneumatics Its Application

Working Principle of Pneumatic System

- Compressed air is generated by the compressor.
- Air flows into the receiver tank for storage.
- Control valves direct airflow to actuators.
- Air pressure causes pistons in cylinders to move.
- Mechanical work is performed (e.g., lifting, pushing).
- Exhausted air is released into the atmosphere.

Basic Components of Pneumatic Systems

- **Compressor**: Generates compressed air.
- **Receiver Tank**: Stores compressed air for steady supply.
- **Control Valves**: Direct and regulate airflow.
- **Actuators**: Convert compressed air into mechanical movement.
- **Filters, Regulators, Lubricators** (FRLs): Ensure air quality and system efficiency.

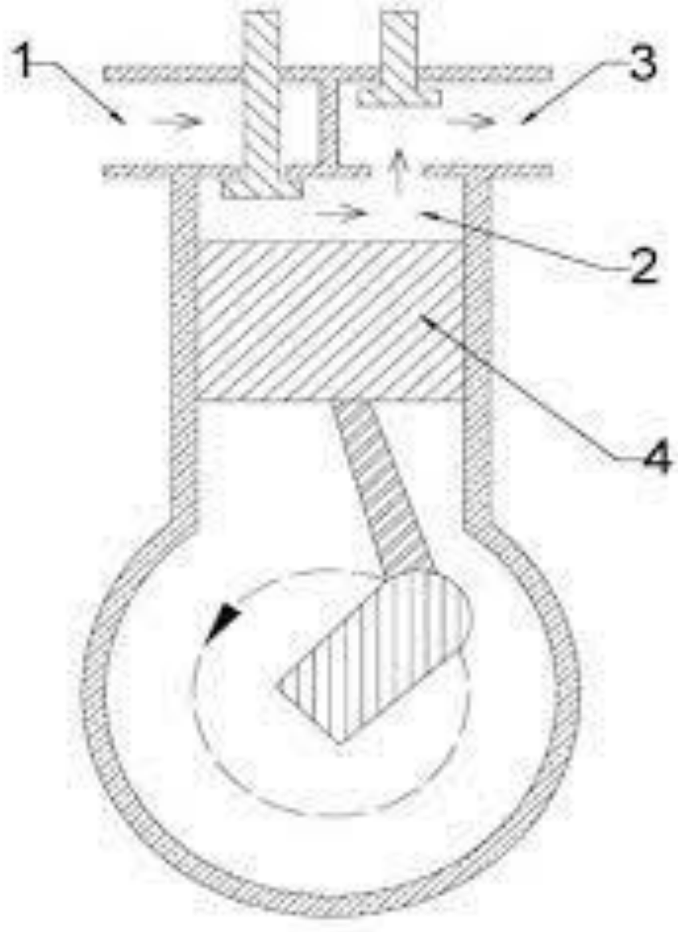
Compressor

Function: Converts ambient air into high-pressure compressed air.

Role: Acts as the power source for the entire system.

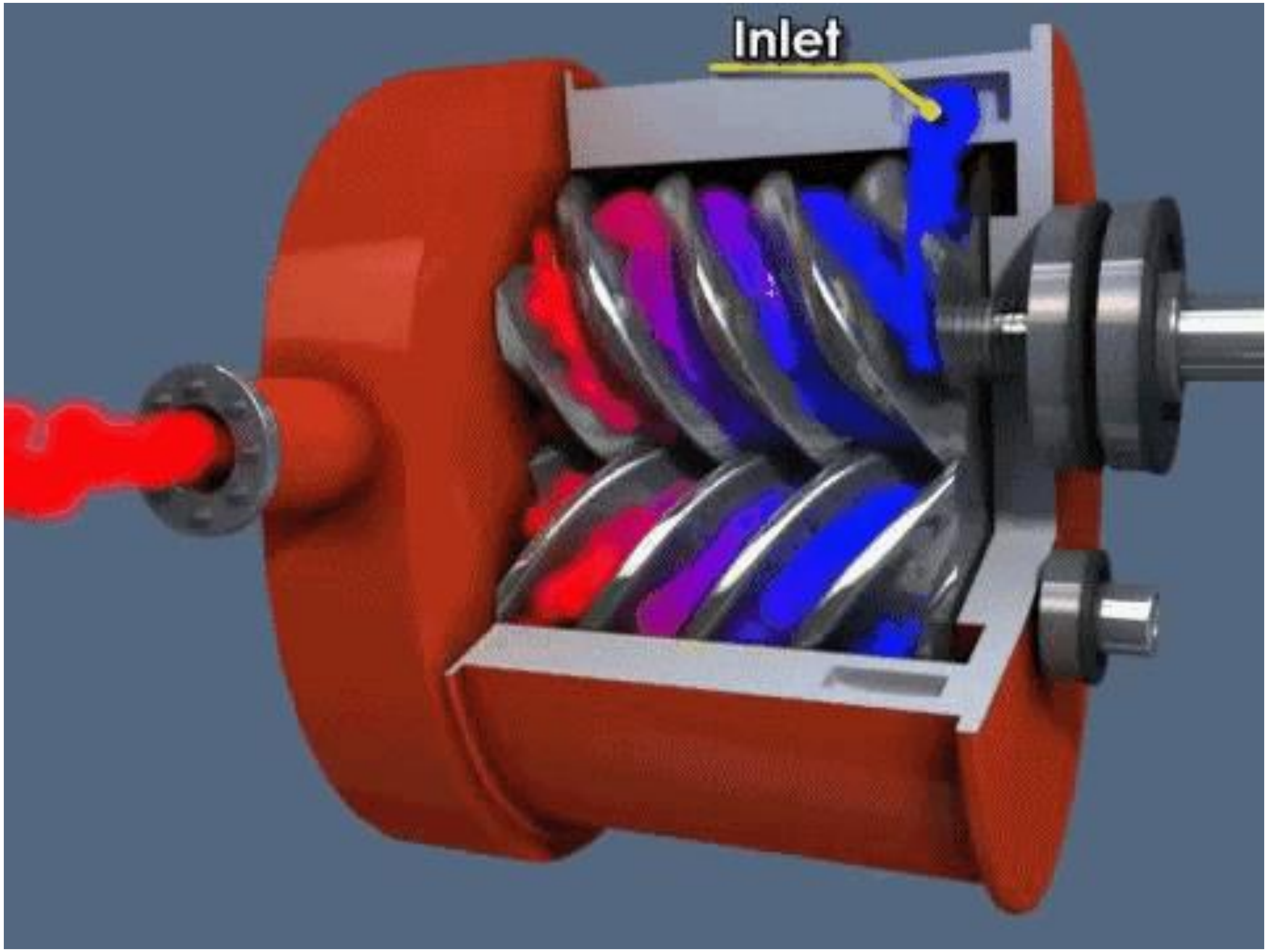
Reciprocating (Piston type)

Uses pistons to compress air.



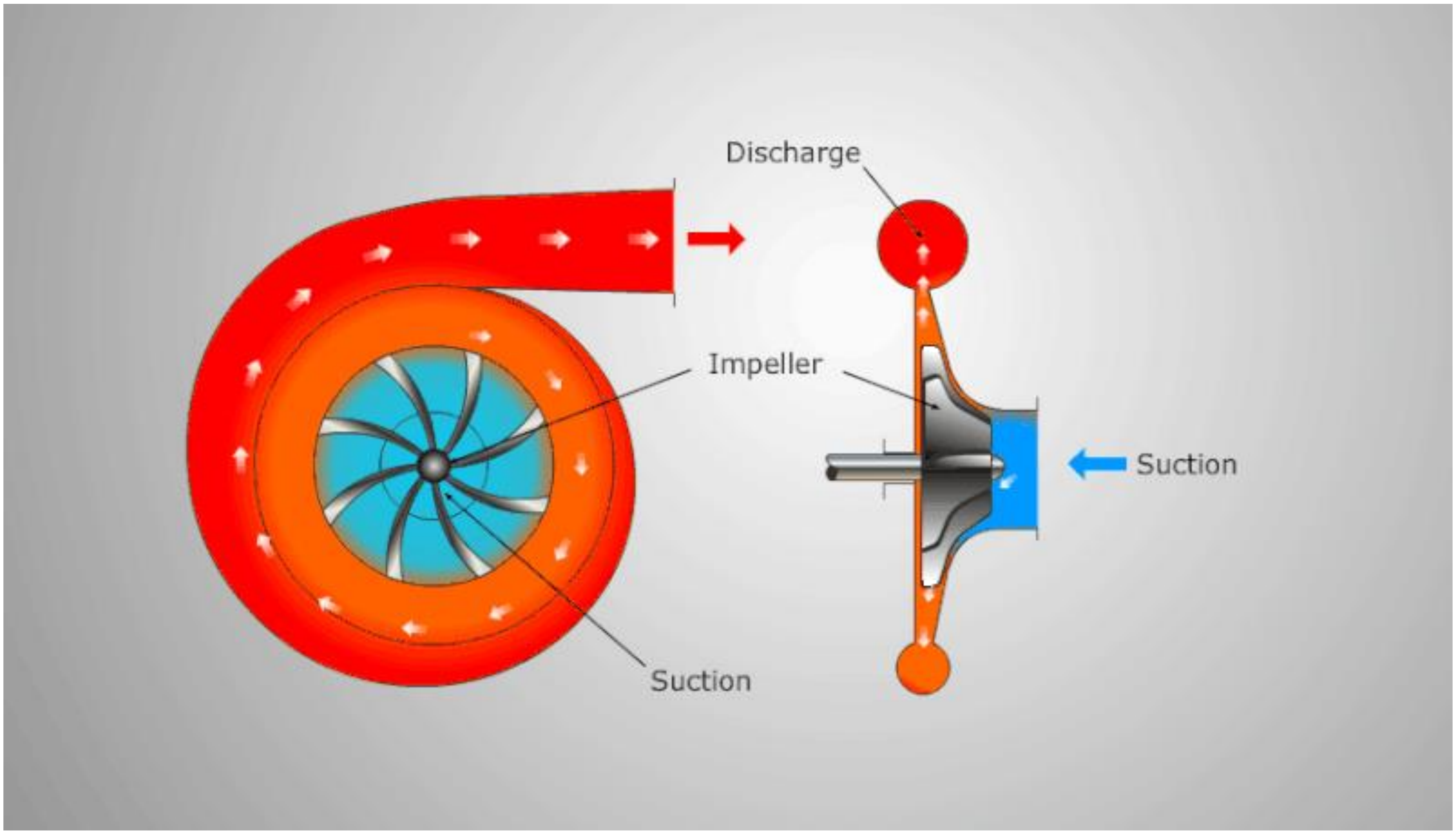
Rotary screw

Uses two meshing screws.



Centrifugal

Uses impellers to accelerate air.

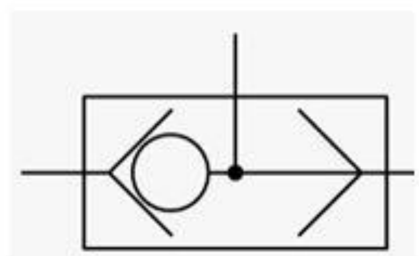


Receiver Tank

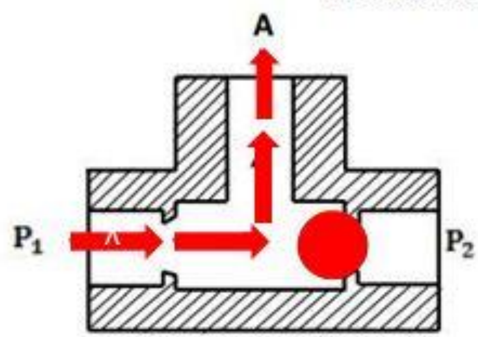
- **Purpose:** Stores compressed air, providing a buffer to meet sudden demand.
- **Benefits:**
  - Reduces cycling of compressor.
  - Maintains steady pressure.
  - Absorbs shocks to system.
- **Placement:** Usually located downstream of the compressor.

Control Valves

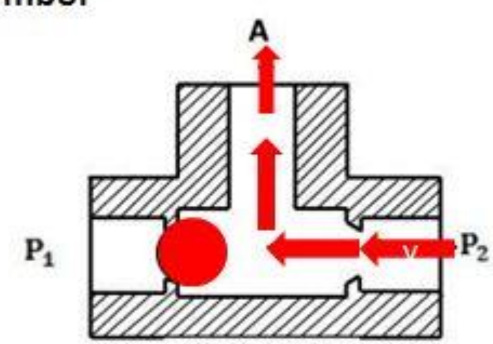
Function: Act as the control center, switching air pathways to operate cylinders and motors.



OR Valve Symbol

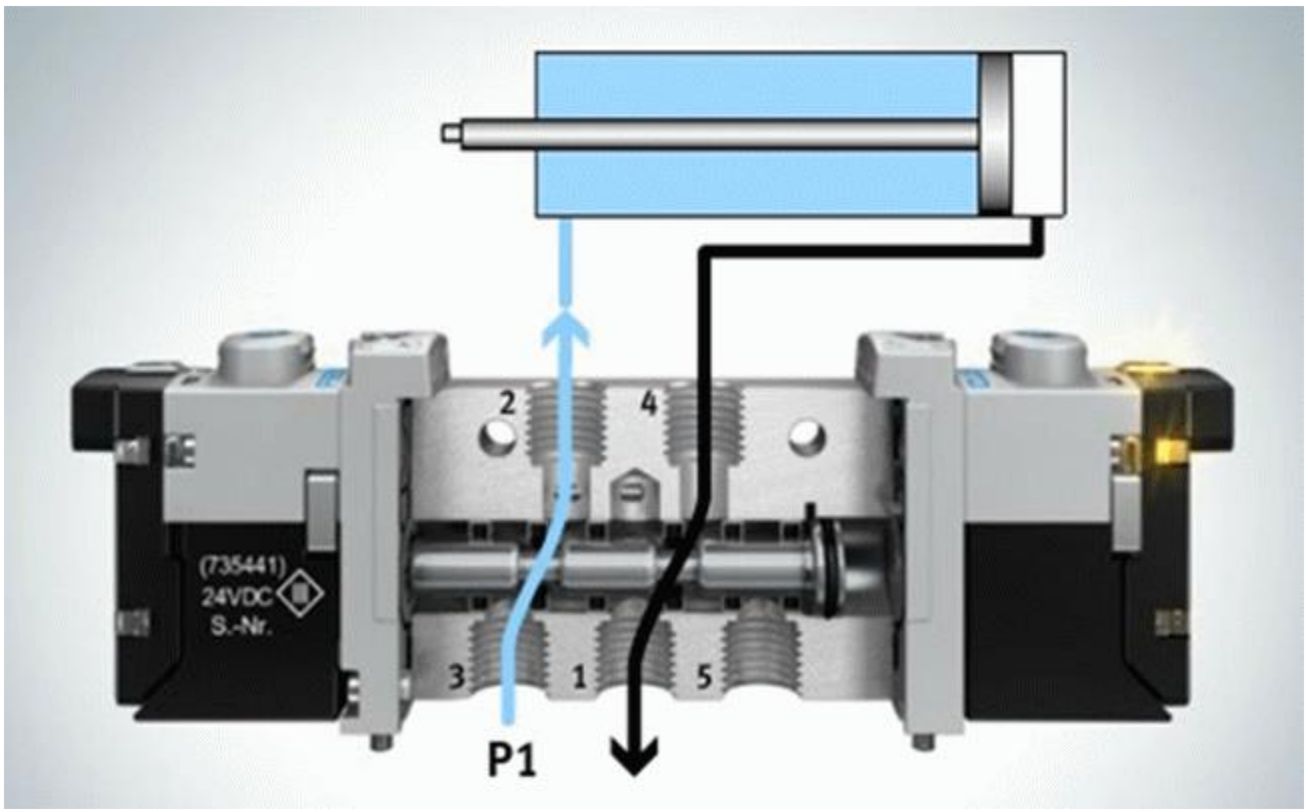


$(a) P_1 > P_2$

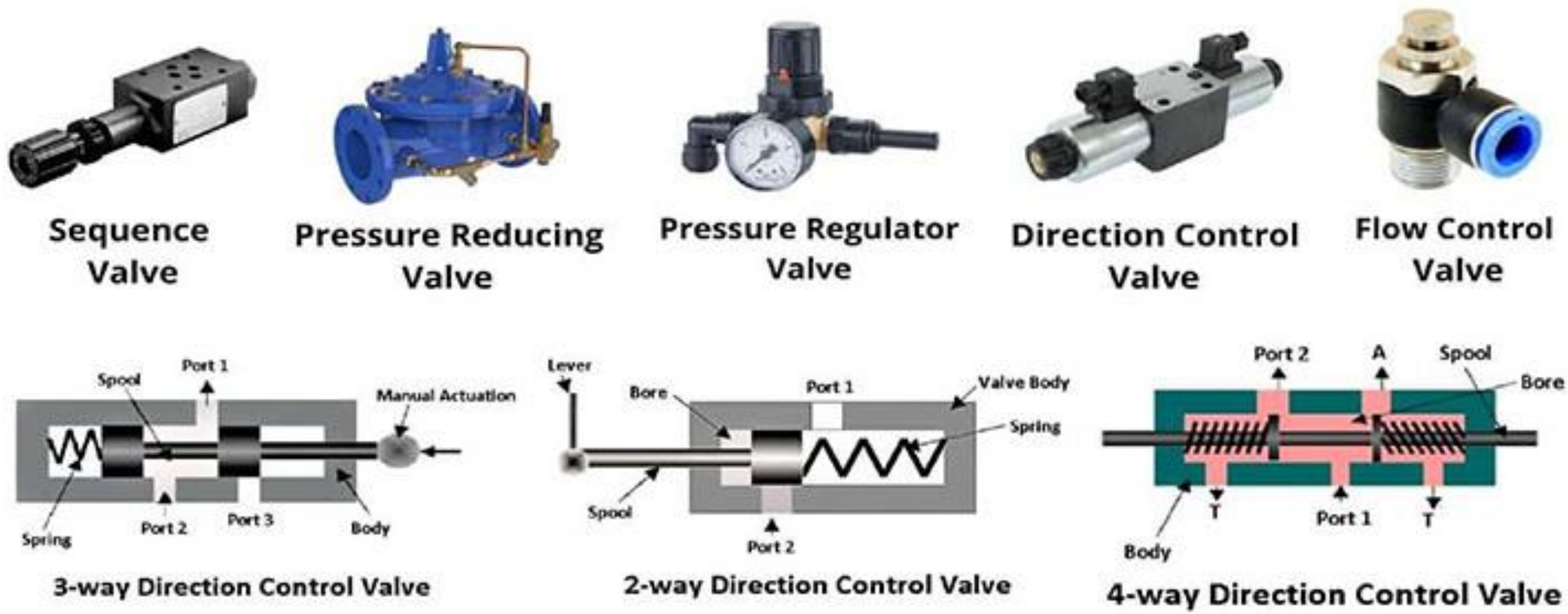


$(b) P_2 > P_1$

Directional Control Valves:
Control the direction of airflow.



Pressure Control Valves:
Regulate system pressure.

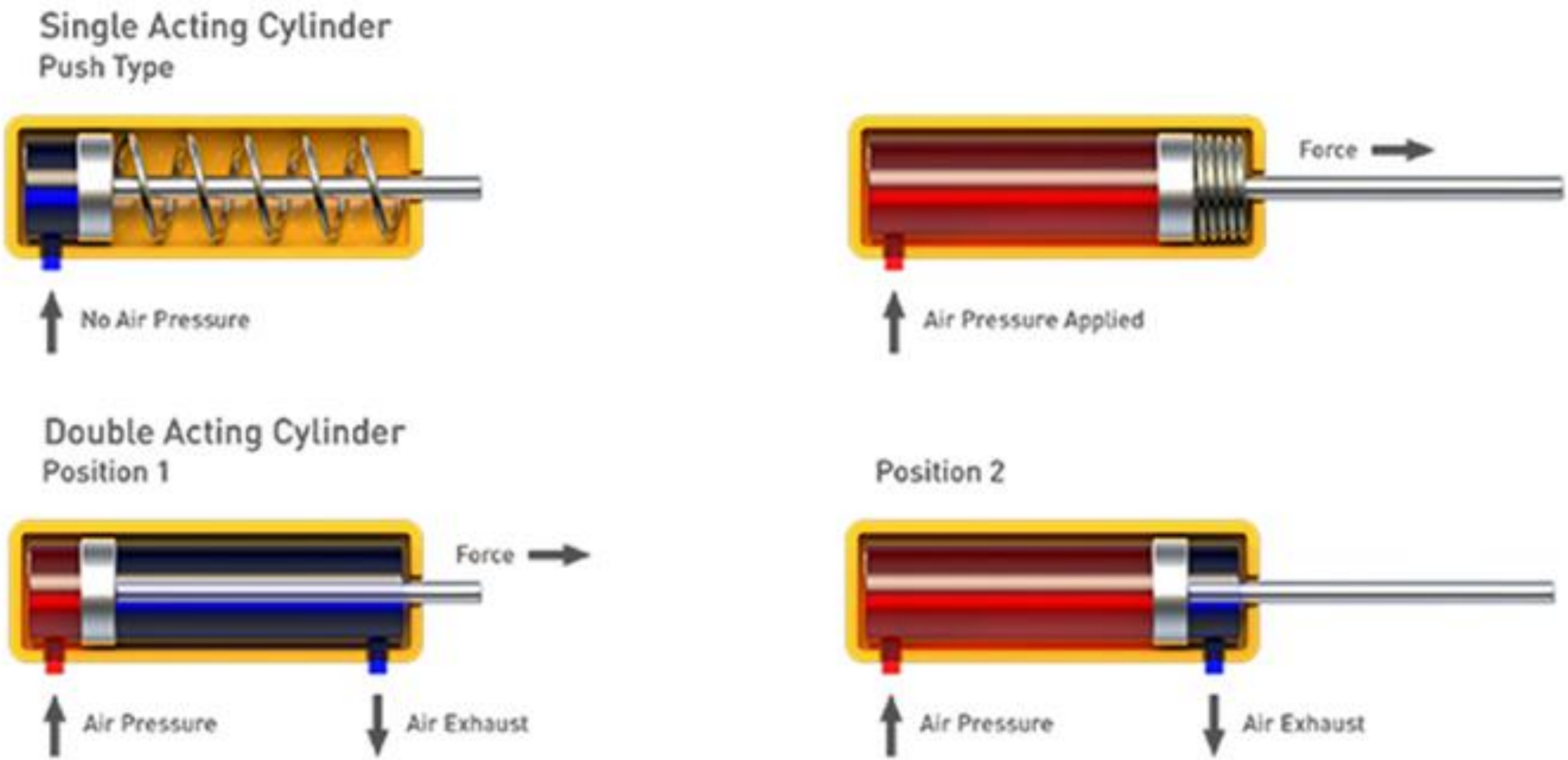


Flow Control Valves:

Control the flow rate of air.

Actuators

Cylinders: Use compressed air to produce linear motion (extension and retraction).

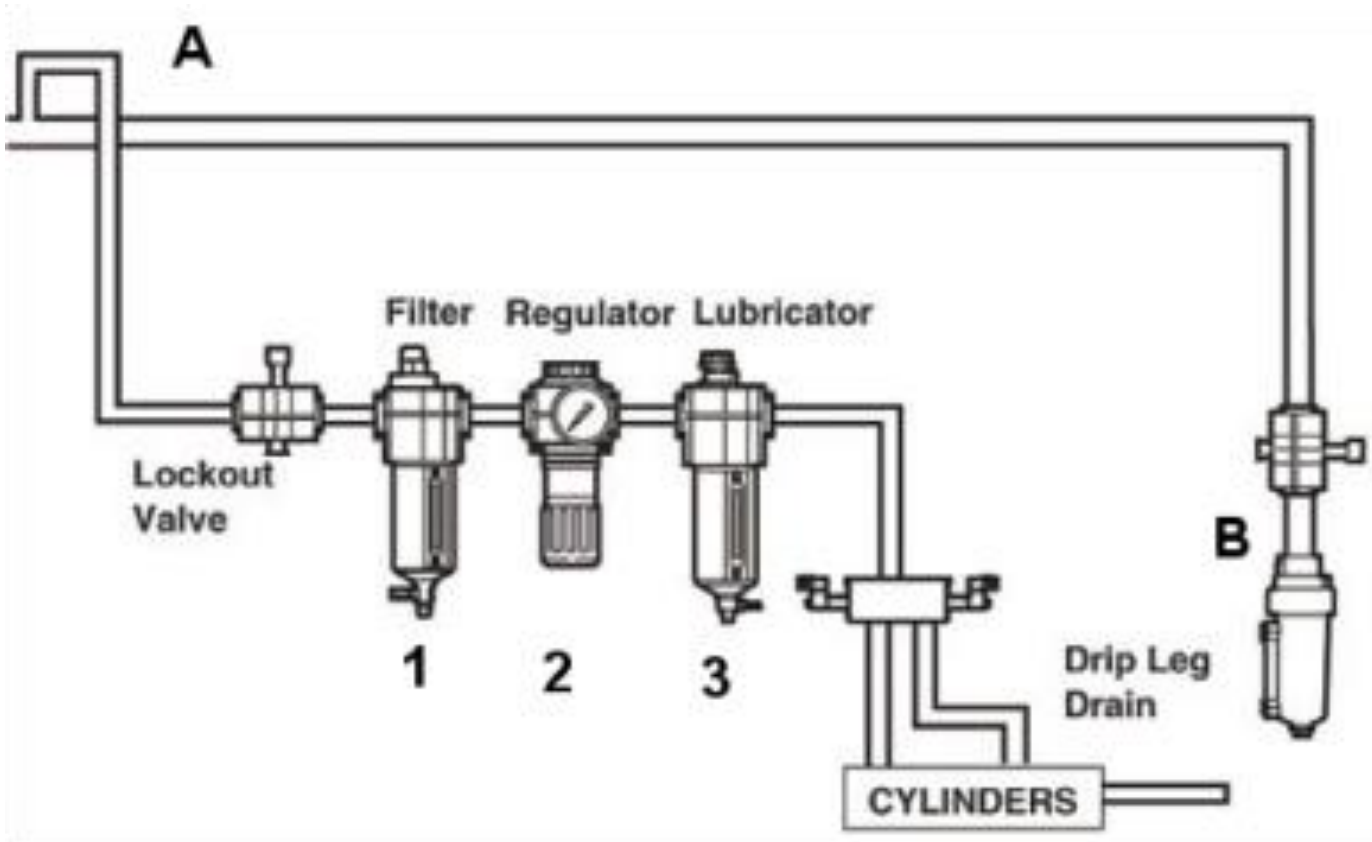


Actuators

- **Pneumatic Motors:** Convert compressed air into rotary motion, used in drills or turntables.

Air pressure acts on piston or rotor, creating mechanical movement

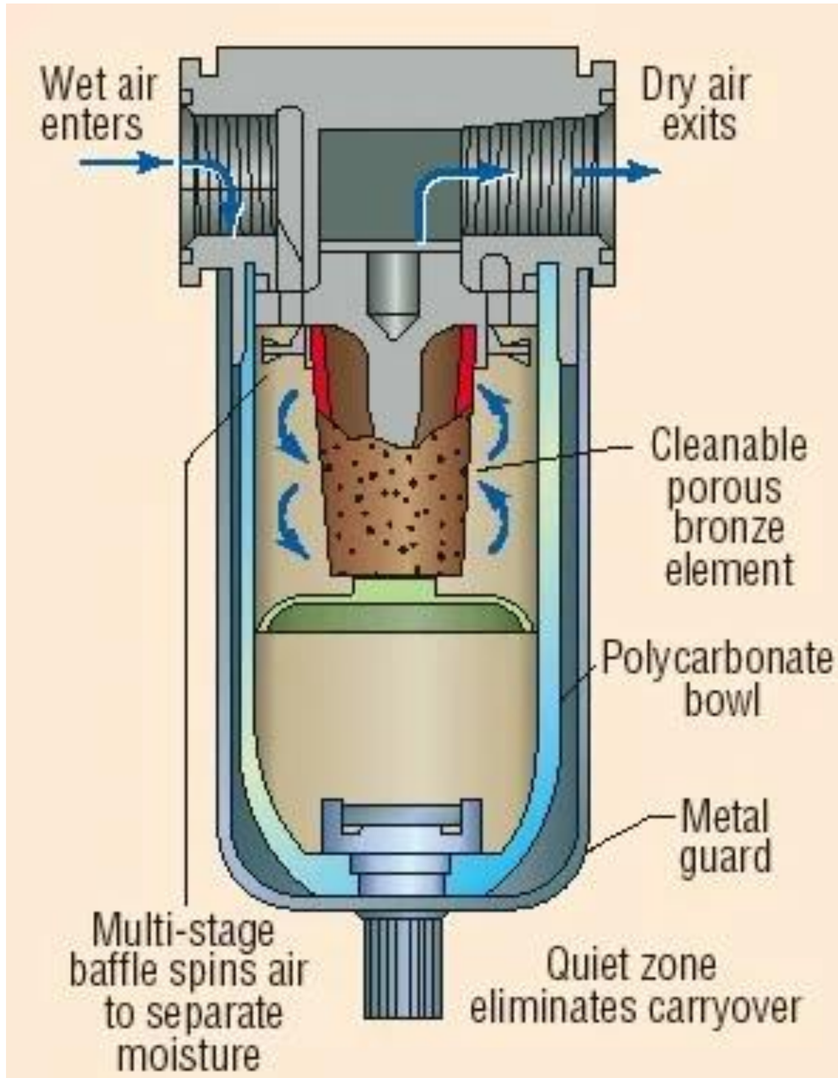
Filters, Regulators, and Lubricators




Clarke
Clarke


SUMAKE

Filters: Remove dirt, water, and oil particles from compressed air.



Regulators: Adjust and maintain consistent pressure.

Lubricators: Add a fine mist of oil to lubricate moving parts, reducing wear.

Importance: Ensures smooth, efficient, and long-lasting operation of the system.

Advantages of Pneumatic Systems

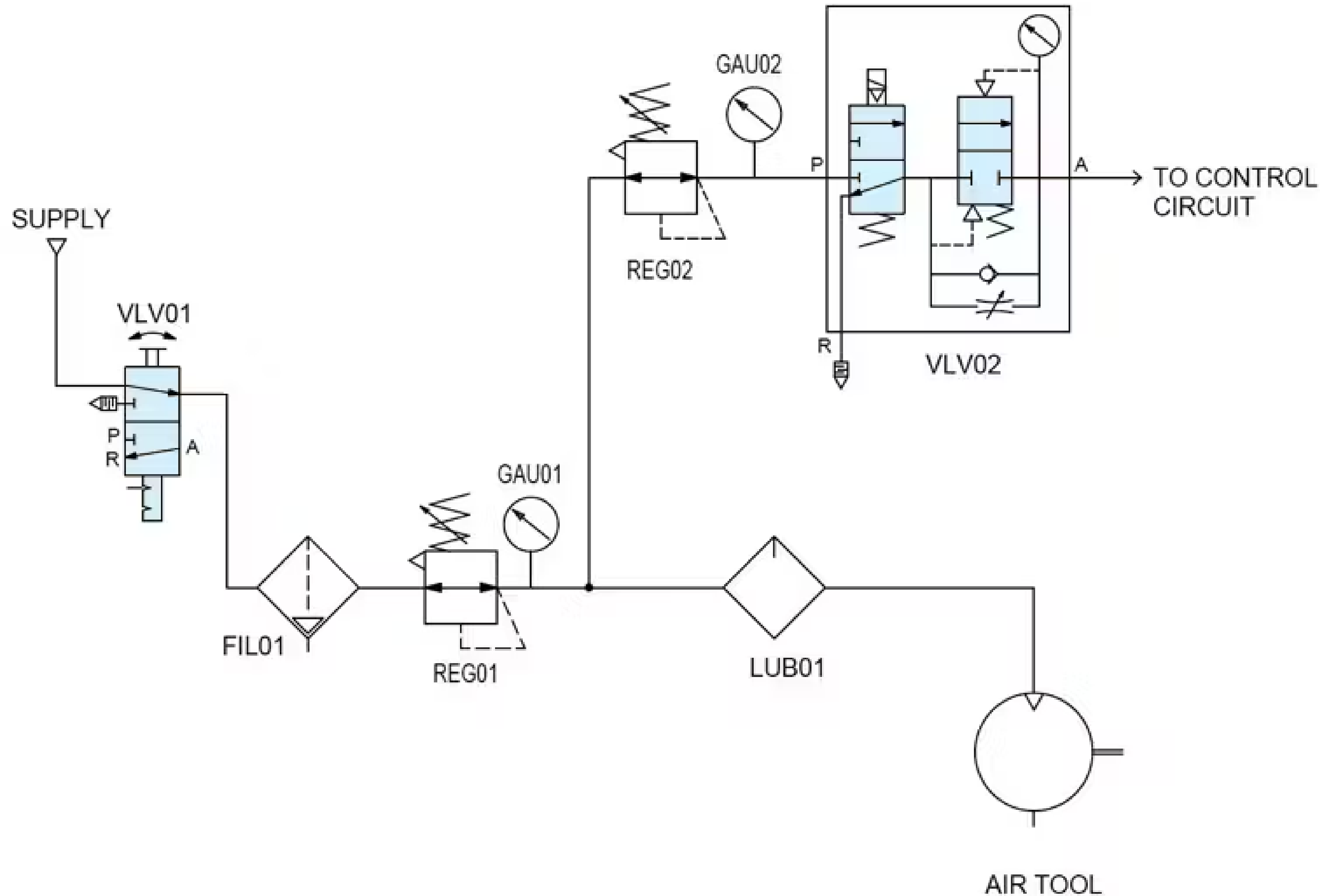
- **Safety:** Non-flammable, non-electrical.
- **Clean:** No oil or grease contamination.
- **Speed:** Rapid movement and response.
- **Simple Maintenance:** Fewer moving parts, easy to repair.
- **Cost-effective:** Low initial and maintenance costs.

Disadvantages of Pneumatic Systems

- **Limited Force:** Less force compared to hydraulic systems.
- **Compressibility:** Air’s compressibility causes less precise control.
- **Noise:** Air exhaust noise can be loud.
- **Energy Loss:** Compression leads to energy wastage.

Applications of Pneumatic Systems

- **Manufacturing:** Automation of assembly lines.
- **Tools:** Air-powered drills, hammers.
- **Transportation:** Brake systems in buses and trucks.
- **Packaging:** Filling, sealing, and stacking machines.
- **Aerospace:** Aircraft control surfaces, landing gear.


SUPPLY
VLV01
P
R
A
FIL01
REG01
GAU01
LUB01
AIR TOOL
REG02
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TO CONTROL
CIRCUIT
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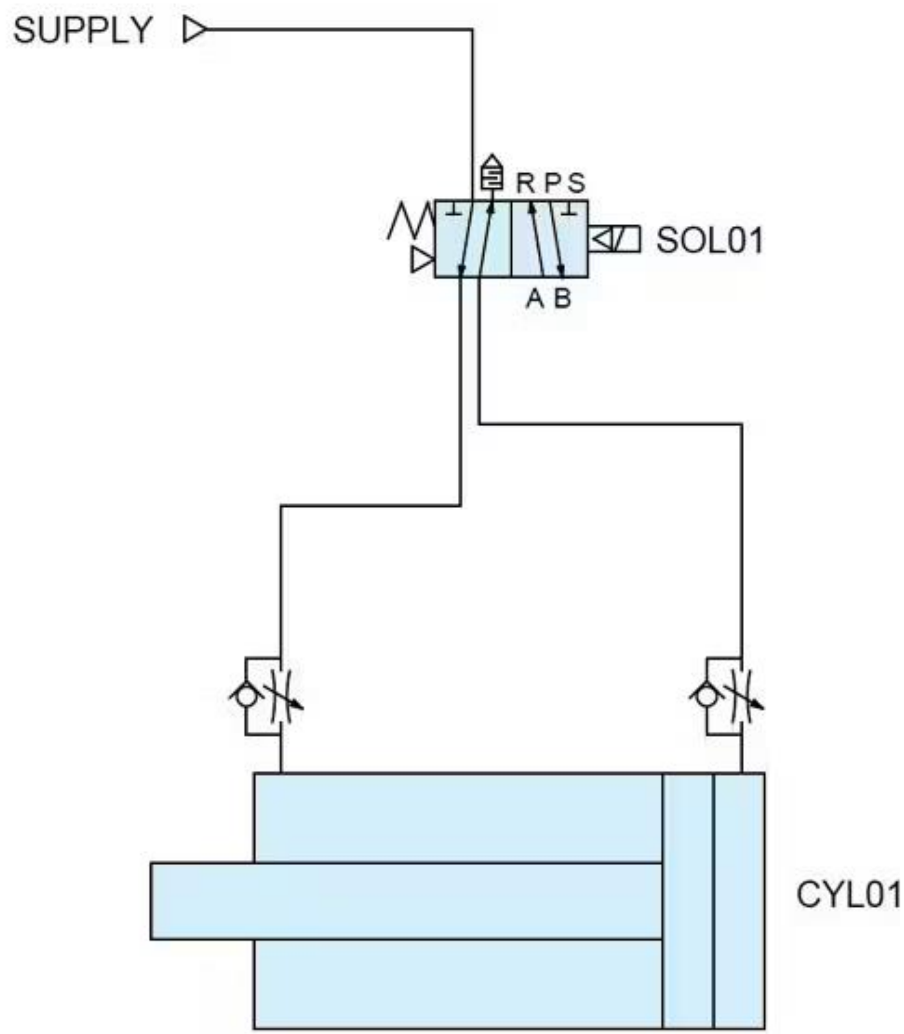
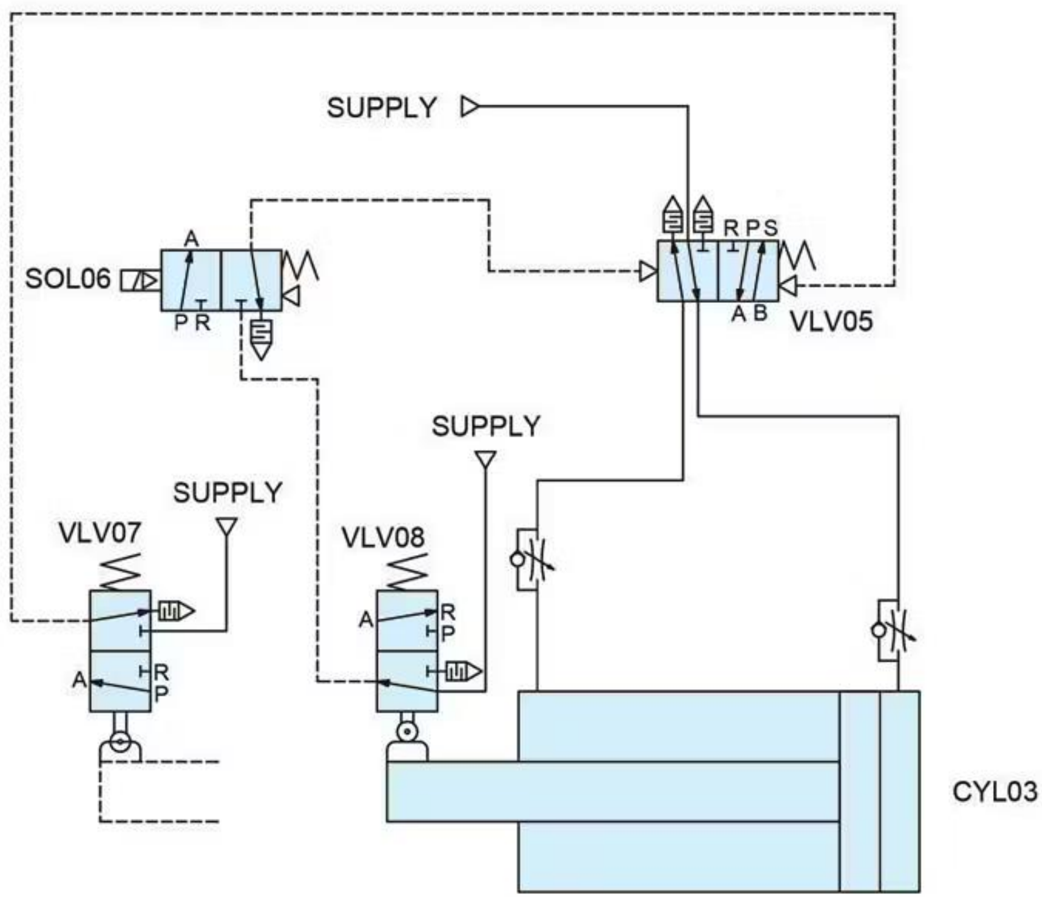
VLV02

Double-Acting Cylinder



Continuous Cycling Cylinder



Pneumatic Systems

Applications

- Used in manufacturing industries
- Automation
- Robotics
- Aircraft systems
- Tools